23|01|2025

* Practice Questions

Q-1 :- Set $A = \{1, 2, 3, 4, 5, 6\}$

$B = \{2, 4, 6, 8\}$

Find out

(i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$ (iv) A'

Sol =

(i) $A \cup B = \{1, 2, 3, 4, 5, 6\} \cup \{2, 4, 6, 8\}$

$A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$

(ii) $A \cap B = \{1, 2, 3, 4, 5, 6\} \cap \{2, 4, 6, 8\}$

$A \cap B = \{2, 4, 6\}$

(iii) $A - B = \{1, 2, 3, 4, 5, 6\} - \{2, 4, 6, 8\}$

$A - B = \{1, 3, 5\}$

(iv) $A' = U - A$

$U = \{1, 2, 3, 4, 5, 6, 8\}$

$A' = \{1, 2, 3, 4, 5, 6, 8\} - \{1, 2, 3, 4, 5, 6\}$

$A' = \{8\}$

Q-2 = Set $A = \{x : x$ is an integer, $1 \leq x \leq 6\}$

$B = \{x : x$ is an even integer, $2 \leq x \leq 8\}$

Convert into Roster form.

Sol =

$A = \{1, 2, 3, 4, 5, 6\}$

$B = \{2, 4, 6, 8\}$ //

Q-3 = $J = \{2, 4, 6, 8, 10\}$

$K = \{3, 6, 9, 12\}$

Find out $J \times K$.

Sol = $J \times K = \{(2,3), (2,6), (2,9), (2,12), (4,3),$
$(4,6), (4,9), (4,12), (6,3), (6,6), (6,9), (6,12),$
$(8,3), (8,6), (8,9), (8,12), (10,3), (10,6), (10,9),$
$(10,12)\}$ //

Q-4 = Set $M = \{x \mid x$ is a prime number less then 20 $\}$

$N = \{x \mid x$ is an odd number less then 10 $\}$

Find out $M \cap N$.

Sol = $M = \{2, 3, 5, 7, 11, 13, 17, 19\}$

$N = \{1, 3, 5, 7, 9\}$

$M \cap N = \{2, 3, 5, 7, 11, 13, 17, 19\} \cap \{1, 3, 5, 7, 9\}$

$M \cap N = \{3, 5, 7\}$ //

Q-5 = Set X = Contain all even number between 1 and 20.

Y = Contain all multiple of 3 between 1 and 20.

Find out :- (i) $X \cap Y$ (ii) $X - Y$

Sol= Set $X = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

$Y = \{3, 6, 9, 12, 15, 18\}$

(i) $X \cap Y = \{2, 4, 6, 8, 10, 12, 14, 16, 18\} \cap \{3, 6, 9, 12, 15, 18\}$

$X \cap Y = \{12, 18\}$ //

(ii) $X - Y = \{2, 4, 6, 8, 10, 12, 14, 16, 18\} - \{3, 6, 9, 12, 15, 18\}$

$X - Y = \{2, 4, 8, 10, 14, 16\}$ //

Q 6 = Set $L = \{x : x$ is a postive integer less then $6\}$

$M = \{x : x$ is a postive integer and multiple of $2\}$

Find out :-

(i) $L \cup M$

(ii) $L \cap M$

Sol= Set $L = \{1, 2, 3, 4, 5\}$

$M = \{2, 4, 6, 8, 10 \ldots\}$

(i) $L \cup M = \{1, 2, 3, 4, 5\} \cup \{2, 4, 6, 8, 10\}$

$L \cup M = \{1, 2, 3, 4, 5, 6, 8, 10\}$ //

(ii) $L \cap M = \{1, 2, 3, 4, 5\} \cap \{2, 4, 6, 8, 10\}$

$L \cap M = \{2, 4\}$ //

(iii) $L - M = \{1, 2, 3, 4, 5\} - \{2, 4, 6, 8, 10\}$

$L - M = \{1, 3, 5\}$ //

(iv) $L^c = U - A$

$U = \{1, 2, 3, 4, 5, 6, 8, 10\}$

$L^c = \{1, 2, 3, 4, 5, 6, 8, 10\} - \{1, 2, 3, 4, 5\}$

$L^c = \{6, 8, 10\}$ //